

GED Math Practice: Measurement (30 Questions)

Instructions

- Answer all 30 questions to the best of your ability.
- Questions are organized by difficulty level: Easy (1–10), Intermediate (11–20), and Challenging (21–30).
- Multiple choice questions should be answered with the letter (A, B, C, or D) that corresponds to the best answer.
- Fill-in-the-blank and short answer questions require you to write your response directly.
- Show your work for all calculations to receive full credit.
- **Recommended time:** 90 minutes total (30 minutes per section).

Easy Questions (1–10)

Question 1: A rectangle has a length of 12 cm and a width of 8 cm. What is the perimeter?

- (A) 20 cm
- (B) 40 cm
- (C) 96 cm
- (D) 60 cm

Question 2: Convert 5 kilometers to meters.

- (A) 50 m
- (B) 500 m
- (C) 5,000 m

(D) 50,000 m

Question 3: A square has a side length of 7 inches. What is its area?

(A) 28 in^2

(B) 49 in^2

(C) 14 in^2

(D) 56 in^2

Question 4: A circle has a radius of 5 cm. What is its circumference? (Use $\pi \approx 3.14$)

(A) 15.7 cm

(B) 31.4 cm

(C) 78.5 cm

(D) 62.8 cm

Question 5: How many millimeters are in 3 centimeters?

(A) 0.3 mm

(B) 30 mm

(C) 300 mm

(D) 3,000 mm

Question 6: A triangle has a base of 10 m and a height of 6 m. What is its area?

(A) 30 m^2

(B) 60 m^2

(C) 16 m^2

(D) 80 m^2

Question 7: Convert 2 hours to minutes.

(A) 60 minutes

(B) 90 minutes

(C) 120 minutes

(D) 180 minutes

Question 8: A rectangular prism has dimensions of $4 \text{ cm} \times 3 \text{ cm} \times 5 \text{ cm}$. What is its volume?

- (A) 12 cm^3
- (B) 60 cm^3
- (C) 20 cm^3
- (D) 47 cm^3

Question 9: How many inches are in 2 feet?

- (A) 12 inches
- (B) 24 inches
- (C) 36 inches
- (D) 48 inches

Question 10: A cylinder has a radius of 3 m and a height of 8 m. What is its volume?
(Use $\pi \approx 3.14$)

- (A) 24 m^3
- (B) 72.96 m^3
- (C) 226.08 m^3
- (D) 452.16 m^3

Intermediate Questions (11–20)

Question 11: A map uses a scale of 1 inch = 50 miles. If the distance on the map is 3 inches, what is the actual distance?

- (A) 50 miles
- (B) 100 miles
- (C) 150 miles
- (D) 200 miles

Question 12: If a car travels 240 miles in 4 hours, what is its average speed?

- (A) 40 mph
- (B) 50 mph
- (C) 60 mph

(D) 80 mph

Question 13: A trapezoid has parallel sides of 8 cm and 12 cm, with a height of 5 cm. What is its area?

(A) 40 cm^2

(B) 50 cm^2

(C) 60 cm^2

(D) 100 cm^2

Question 14: Convert 3,500 grams to kilograms.

(A) 0.35 kg

(B) 3.5 kg

(C) 35 kg

(D) 350 kg

Question 15: A rectangular garden is 15 m long and 10 m wide. What is its perimeter?

(A) 25 m

(B) 50 m

(C) 75 m

(D) 150 m

Question 16: The ratio of boys to girls in a class is 3:2. If there are 12 boys, how many girls are there?

(A) 6 girls

(B) 8 girls

(C) 10 girls

(D) 12 girls

Question 17: A cone has a radius of 4 cm and a height of 9 cm. What is its volume? (Use $\pi \approx 3.14$)

(A) 36 cm^3

(B) 72 cm^3

(C) 150.72 cm^3

(D) 301.44 cm^3

Question 18: Convert 5 yards to feet.

(A) 5 feet

(B) 10 feet

(C) 15 feet

(D) 20 feet

Question 19: A sphere has a radius of 6 cm. What is its volume? (Use $\pi \approx 3.14$)

(A) 113.04 cm^3

(B) 452.16 cm^3

(C) 904.32 cm^3

(D) $1,131.36 \text{ cm}^3$

Question 20: A cube has a side length of 4 inches. What is its surface area?

(A) 16 in^2

(B) 48 in^2

(C) 64 in^2

(D) 96 in^2

Challenging Questions (21–30)

Question 21: A rectangular prism has dimensions of $5 \text{ cm} \times 6 \text{ cm} \times 7 \text{ cm}$. What is its surface area?

(A) 107 cm^2

(B) 214 cm^2

(C) 210 cm^2

(D) 420 cm^2

Question 22: If a worker can complete a job in 8 hours working at a constant rate, how long would it take 2 workers to complete the same job?

(A) 2 hours

- (B) 4 hours
- (C) 8 hours
- (D) 16 hours

Question 23: A circle has a diameter of 14 cm. What is its area? (Use $\pi \approx 3.14$)

- (A) 43.96 cm²
- (B) 87.92 cm²
- (C) 153.86 cm²
- (D) 615.44 cm²

Question 24: A scale drawing of a building uses a scale of 1:100. If the height on the drawing is 5 cm, what is the actual height of the building?

- (A) 5 m
- (B) 50 m
- (C) 100 m
- (D) 500 m

Question 25: Convert 72 inches to yards.

- (A) 2 yards
- (B) 3 yards
- (C) 6 yards
- (D) 12 yards

Question 26: A triangular prism has a triangular base with an area of 20 cm² and a height (length) of 10 cm. What is its volume?

- (A) 30 cm³
- (B) 100 cm³
- (C) 200 cm³
- (D) 400 cm³

Question 27: If the ratio of the sides of two similar rectangles is 2:3, what is the ratio of their areas?

- (A) 2:3

- (B) 4:9
- (C) 6:9
- (D) 8:27

Question 28: A compound shape consists of a rectangle ($8 \text{ m} \times 5 \text{ m}$) with a triangle on top (base 8 m, height 3 m). What is the total area?

- (A) 40 m^2
- (B) 52 m^2
- (C) 60 m^2
- (D) 72 m^2

Question 29: A hexagon has a perimeter of 48 cm. What is the length of each side?

- (A) 6 cm
- (B) 8 cm
- (C) 12 cm
- (D) 24 cm

Question 30: A cylindrical tank has a radius of 2 m and a height of 10 m. How much liquid can it hold? (Use $\pi \approx 3.14$)

- (A) 62.8 m^3
- (B) 125.6 m^3
- (C) 251.2 m^3
- (D) $1,256 \text{ m}^3$

Answer Key

Study Tips

- **Master unit conversions:** Practice converting between metric and imperial units regularly. Create flashcards for common conversions (cm to m, feet to yards, etc.).

- **Memorize key formulas:** Know the formulas for area, perimeter, volume, and circumference by heart. Understanding what each formula represents helps you apply it correctly.
- **Draw diagrams:** Always sketch the problem when dealing with geometry. Visual representations help you understand the problem better and reduce calculation errors.
- **Check your work:** After solving a problem, verify your answer by using a different method or plugging the answer back into the original equation.
- **Practice with real-world contexts:** Measurement problems often relate to practical situations. Connect abstract problems to real objects to deepen your understanding and retention.

Question	Answer	Explanation
1	B	Perimeter of rectangle = $2(l + w) = 2(12 + 8) = 40$ cm
2	C	1 km = 1,000 m; therefore 5 km = 5,000 m
3	B	Area of square = $s^2 = 7^2 = 49$ in ²
4	B	Circumference = $2\pi r = 2(3.14)(5) = 31.4$ cm
5	B	1 cm = 10 mm; therefore 3 cm = 30 mm
6	A	Area of triangle = $\frac{1}{2}bh = \frac{1}{2}(10)(6) = 30$ m ²
7	C	1 hour = 60 minutes; 2 hours = 120 minutes
8	B	Volume = $l \times w \times h = 4 \times 3 \times 5 = 60$ cm ³
9	B	1 foot = 12 inches; 2 feet = 24 inches
10	C	Volume of cylinder = $\pi r^2 h = 3.14(3^2)(8) = 226.08$ m ³
11	C	3 in $\times$ 50 mi/in = 150 miles
12	C	Speed = $\frac{\text{distance}}{\text{time}} = \frac{240}{4} = 60$ mph
13	B	Area of trapezoid = $\frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(8 + 12)(5) = 50$ cm ²
14	B	1 kg = 1,000 g; 3,500 g = 3.5 kg
15	B	Perimeter = $2(l + w) = 2(15 + 10) = 50$ m
16	B	Ratio is 3:2; if boys = 12, then $12/3 = 4$ units; girls = $4 \times 2 = 8$
17	C	Volume of cone = $\frac{1}{3}\pi r^2 h = \frac{1}{3}(3.14)(4^2)(9) = 150.72$ cm ³
18	C	1 yard = 3 feet; 5 yards = 15 feet
19	C	Volume of sphere = $\frac{4}{3}\pi r^3 = \frac{4}{3}(3.14)(6^3) = 904.32$ cm ³
20	D	Surface area of cube = $6s^2 = 6(4^2) = 96$ in ²
21	B	SA = $2(lw + lh + wh) = 2(5 \cdot 6 + 5 \cdot 7 + 6 \cdot 7) = 2(107) = 214$ cm ²
22	B	With 2 workers, time is halved: $8/2 = 4$ hours
23	C	Area = $\pi r^2 = 3.14(7^2) = 153.86$ cm ² (radius = $14/2 = 7$)
24	A	5 cm $\times$ 100 = 500 cm = 5 m
25	A	1 yard = 36 inches; 72 inches = $72/36 = 2$ yards
26	C	Volume of prism = base area $\times$ height = $20 \times 10 = 200$ cm ³
27	B	If linear ratio is 2:3, area ratio is $(2 : 3)^2 = 4 : 9$
28	B	Rectangle: $8 \times 5 = 40$ m ² ; Triangle: $\frac{1}{2}(8)(3) = 12$ m ² ; Total = 52 m ²
29	B	Regular hexagon has 6 equal sides; $48/6 = 8$ cm
30	B	Volume = $\pi r^2 h = 3.14(2^2)(10) = 125.6$ m ³